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Symmetry-Based Formation Control on Cycle Graphs Using Dihedral Point Groups ⋆

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Abstract: This work develops a symmetry-based framework for formation control on cycle graphs using Dihedral point-group constraints. We show that enforcing inter-agent reflection symmetries, together with anchoring a single designated agent to its prescribed mirror axis, is sufficient to realize every $\mathcal{C}_{nv}$-symmetric configuration using only $n-1$ communication links. The resulting control laws have a matrix-weighted Laplacian structure and guarantee exponential convergence to the desired symmetric configuration. Simulation results are provided to support the theoretical analysis.

Keywords: Formation Control, Multi-agent systems, Symmetry

1. INTRODUCTION

The study of distributed formation control in multi-agent systems (MAS) has drawn increasing attention in recent years. It plays a central role in many application domains, including distributed sensing architectures (Wu et al. (2024)), motion coordination for autonomous vehicle platoons (Dai et al. (2018); Chung et al. (2018)), and swarm coordination for mapping, surveillance, and localization (Tron et al. (2016)). A fundamental requirement to these settings is the design of control strategies that steer the agents to a prescribed spatial configuration through purely local interactions. A key challenge, however, lies in balancing sparse information exchange with guaranteed convergence to the desired configuration.

Classical approaches to this problem include distance-based schemes (Krick et al. (2009); de Queiroz et al. (2019)) and bearing-based schemes (Zhao and Zelazo (2019)), both of which leverage results from rigidity theory, relying on minimal infinitesimal rigidity (MIR) as a crucial structural requirement to guarantee convergence to a desired formation shape. In $\mathbb{R}^2$, MIR requires $2n-3$ communication links to achieve a formation of n agents.

A parallel line of research has explored augmented Laplacian formulations, where the inter-agent geometric relations are encoded via structured weights (Trinh et al. (2018)), including complex-Laplacian approaches (Garcia de Marina (2021); Zhou et al. (2025)) in which the scalar entries are replaced with complex numbers, naturally capturing rotations, translations, and scaling. *Matrix-weighted Laplacian* formulations, where the weights are structured matrices encoding inter-agent relations, offer another approach. Examples include projection-matrix weights seen in bearing-based control (Zhao and Zelazo (2019)), and matrix-structured weights that mimic the role of complex weights (Fan et al. (2025)).

Within this research field, spatial *symmetries* offer a potential alternative to rigidity-based approaches (Zelazo et al. (2025); Martinez and Zelazo (2026)). In many practical formations, the geometric relationships between agents naturally appear as rotations or reflections (about a common centroid or axis), often arising from sensing coverage or communication requirements. By exploiting these relationships, traditional geometric constraints (e.g., distances or bearings) can be replaced with point group relations, significantly reducing the number of required inter-agent communication links. Recent results (Martinez and Zelazo (2026)) demonstrate the potential of this approach as a structured matrix-weighted Laplacian, showing that rotational symmetry constraints can drive the system to a unique symmetric configuration while using only $n-1$ inter-agent communication links in $\mathbb{R}^2$, matching the minimal connectivity required between the agents.

Planar formations frequently exhibit dihedral symmetries, which combine an n-fold rotational symmetry with a collection of reflection symmetries. While the rotational component has been addressed in prior work (Martinez and Zelazo (2026)), the reflectional component introduces a more challenging problem. Enforcing mirror constraints alone may produce large families of admissible configurations, leading to symmetric flexes. On the other hand, enforcing all symmetry relations in the formation is generally redundant, since many of these relations are implied by the symmetric structure. As a consequence, designing a complete symmetry-based framework for formations exhibiting dihedral symmetries has yet to be established.

This work extends the symmetry-based framework in $\mathbb{R}^2$ to formations exhibiting dihedral symmetries by exploiting the properties of reflection constraints. We show that although inter-agent reflection constraints alone may lead to ambiguous configurations, introducing a leader-follower architecture resolves this ambiguity and suffices to solve the formation control problem using *only* $n-1$ communica-

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tion links. We provide a stability and convergence analysis of the resulting control law.

Notations A (finite, simple) graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of two non-empty sets, the set of vertices (nodes) $\mathcal{V} = \{1, ..., n\}$, and the set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, where each edge $ij \in \mathcal{E}$ connects two distinct vertices $i, j \in \mathcal{V}$. Throughout, $\mathcal{G}$ is assumed to be undirected. The graph Laplacian of $\mathcal{G}$ is defined as $L(\mathcal{G}) = EE^T$, where $E \in \mathbb{R}^{n\times m}$ is the incidence matrix (using an arbitrary edge orientation). A path is a sequence of distinct vertices from i to j such that consecutive vertices are adjacent. Let $I_n \in \mathbb{R}^{n\times n}$ and $\mathbb{1}_n \in \mathbb{R}^n$ denote the identity matrix and all-one column vector of dimension n respectively, and e_i denote the i-th standard basis vector in $\mathbb{R}^n$. The Kronecker product is denoted $\otimes$.

2. PRELIMINARIES

This section reviews the key notions from graph theory and group theory that will be used throughout the paper.

2.1 Symmetry in Graphs

Group theory provides a fundamental mathematical framework for describing symmetries. The basic definitions relevant to this work are recalled below.

Definition 1. A *group* is a set Γ with a binary operation $\circ : \Gamma\times\Gamma \to \Gamma$ that is associative, admits an identity $\mathrm{id} \in \Gamma$, and for which every $a \in \Gamma$ has an inverse $a^{-1} \in \Gamma$ satisfying $a\circ a^{-1} = a^{-1}\circ a = \mathrm{id}$. The *order* of Γ is $|\Gamma|$. A subset $B \subseteq \Gamma$ that is closed under $\circ$ and forms a group is a *subgroup*.

Definition 2. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite, simple graph. An *automorphism* of $\mathcal{G}$ is a permutation $\psi : \mathcal{V} \to \mathcal{V}$ such that $uv \in \mathcal{E} \Leftrightarrow \psi(u)\psi(v) \in \mathcal{E}$.

One can express every permutation as a composition of disjoint cycles of the permutation. A cycle is a successive action of the permutation that sends a vertex back to itself, i.e., $i \to \psi(i) \to \psi(\psi(i)) \to \cdots \to \psi^k(i) = i$, where $\psi^k = \underbrace{\psi\circ\cdots\circ\psi}_{k \text{ times}}$. Such a cycle is compactly written using the *cycle notation*, denoted by $(i\,\psi(i)\cdots\psi^{k-1}(i))$. The integer k is the *length* of the cycle.

In the context of graphs, the set of all automorphisms of $\mathcal{G}$ forms the group $\mathrm{Aut}(\mathcal{G})$ under composition. This group captures the *symmetries* of the graph: each element of $\mathrm{Aut}(\mathcal{G})$ is a relabeling of the vertices that preserves the adjacency structure. Often, one works not with the full automorphism group but with a chosen subgroup $\Gamma \subseteq \mathrm{Aut}(\mathcal{G})$. In this case, we say informally that "Γ acts as symmetries of the graph," or simply that $\mathcal{G}$ is Γ-symmetric. Different choices of subgroups capture distinct types of structural symmetries.

2.2 Symmetry in Frameworks

When dealing with formation control problems, the embedding of symmetric graphs in Euclidean space is of particular interest. In this direction, we now consider *symmetry of frameworks* (Schulze and Whiteley (2017)). A framework in $\mathbb{R}^2$ is defined as the pair $(\mathcal{G}, p)$, where $p : \mathcal{V} \to \mathbb{R}^2$ assigns each vertex $i \in \mathcal{V}$ a position $p_i \in \mathbb{R}^2$ representing the physical location of the agents. We denote the configuration of the entire network as the stacked vector $p = \begin{bmatrix} p_1^T & \cdots & p_n^T \end{bmatrix}^T$.

We associate each automorphism of the graph with a corresponding Euclidean isometry (a rotation or reflection) through an appropriate group representation.

Definition 3. Let $\Gamma \subseteq \mathrm{Aut}(\mathcal{G})$, and $\tau : \Gamma \to O(\mathbb{R}^2)$ be a group homomorphism giving an orthogonal representation of Γ on $\mathbb{R}^2$. A framework $(\mathcal{G}, p)$, with $p \in \mathbb{R}^2$, is called $\tau(\Gamma)$-*symmetric* if

$$\tau(\gamma)p_i = p_{\gamma(i)} \quad \forall\gamma \in \Gamma, \quad i \in \mathcal{V}. \tag{1}$$

We use the standard Schoenflies notation for point groups (Atkins et al. (1970); Altmann and Herzig (1994)). In $\mathbb{R}^2$, the possible point groups are

i) *Rotational groups* $\mathcal{C}_n$: the cyclic group generated by an n-fold rotation about the origin.
ii) *Reflection groups* $\mathcal{C}_s$: a point group consisting of a single mirror reflection across a fixed line in $\mathbb{R}^2$.
iii) *Dihedral groups* $\mathcal{C}_{nv}$: the point group of order $2n$, generated by $\mathcal{C}_n$ together with n distinct reflections $\{\sigma_1, \ldots, \sigma_n\}$ whose mirror lines are spaced by angles of $2\pi/n$.

The rotational subgroup $\mathcal{C}_n$ and each reflection subgroup $\mathcal{C}_s^{(k)} := \{\mathrm{id}, \sigma_k\}$ embed naturally in $\mathcal{C}_{nv}$, i.e., $\mathcal{C}_n \subset \mathcal{C}_{nv}$, and $\mathcal{C}_s^{(k)} \subset \mathcal{C}_{nv}$ (for $k = 1, \ldots, n$). The reflection automorphisms ($\mathcal{C}_s^{(k)}$) of the cycle graph C_n naturally divide into two geometrically distinct classes, and we introduce the notation $\mathscr{F}_s$ and $\mathscr{S}_s$ to distinguish between them:

i) *Free reflections*:
$$\mathscr{F}_s := \{\sigma \in \mathcal{C}_{nv} : \sigma(i) \neq i \text{ for all } i \in \mathcal{V}\}.$$
ii) *Self reflections*:
$$\mathscr{S}_s := \{\sigma \in \mathcal{C}_{nv} : \sigma(i) = i \text{ for some } i \in \mathcal{V}\}.$$

These sets form a disjoint partition of the reflections of $\mathcal{C}_{nv}$:
$$\mathscr{R}_s := \{\sigma_1, \ldots, \sigma_n\} = \mathscr{F}_s \cup \mathscr{S}_s.$$

Free reflections encode mirror relations between distinct agents, while self reflections constrain agents to lie on specific mirror axes. Both types are required to realize the full dihedral symmetry $\mathcal{C}_{nv}$.

By considering the graphs in Fig. 1 as planar frameworks $(\mathcal{G}, p)$, their geometric symmetry depends entirely on the embedding. Depending on the embedding, the framework may exhibit the full dihedral symmetry $\mathcal{C}_{4v}$ (Fig. 1(a)), only the rotational symmetry $\mathcal{C}_4$ (Fig. 1(b)), a single reflection symmetry $\mathcal{C}_s$ (Fig. 1(c)) or no nontrivial symmetry at all (Fig. 1(d)).

As outlined in Definition 3, the geometric action of the point groups can be represented through a homomorphism $\tau : \Gamma \to O(\mathbb{R}^2)$. For any two vertices $i, j \in \mathcal{V}$, we denote by $\gamma_{ij} \in \mathcal{C}_n$ the rotational group element satisfying (1), i.e., the automorphism that maps $i \mapsto j$. The representation of the element $\tau(\gamma_{ij})$ is given as the standard planar rotation by an angle $\theta = 2\pi/n$,

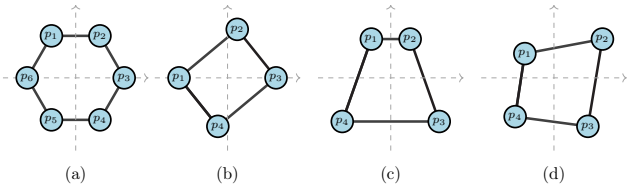


Fig. 1. Symmetric frameworks with C_n as the underlying graph. (a) A $\mathcal{C}_{6v}$-symmetric graph, (b) has a 180° rotation symmetry ($\mathcal{C}_2$), (c) a framework with a reflective symmetry ($\mathcal{C}_s$) about the y-axis, and (d) a framework with no nontrivial symmetry.

$$\tau(\gamma_{ij}) = R(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix},$$

and the inverse relation satisfies $\tau(\gamma_{ji}) = R(-\theta) = R(\theta)^T = \tau(\gamma_{ij})^T$. We also use the notation $\gamma_{ij} \in \mathscr{F}_s$ for reflection-induced automorphisms mapping $i \mapsto j$. Let $\mathcal{L}_{ij}$ denote the corresponding mirror line, and $\hat{n}_{ij}$ its unit normal vector. The representation of γ_{ij} is given by the Householder matrix

$$\tau(\gamma_{ij}) = H(\hat{n}_{ij}) = I_2 - 2\,\hat{n}_{ij}\hat{n}_{ij}^\top.$$

This matrix satisfies $\tau(\gamma_{ij})^2 = I_2$, $\tau(\gamma_{ij}) = \tau(\gamma_{ji})$, and $\det(\tau(\gamma_{ij})) = -1$, as expected for a reflection. Similarly, for a self reflection we denote $\gamma_i \in \mathscr{S}_s$, with mirror line $\mathcal{L}_i$ and normal vector $\hat{n}_i$.

3. SYMMETRY-BASED FORMATION CONTROL FOR C_n

In this section we introduce the symmetry-based formation control problem, specialized for the cycle graph C_n. We choose the cycle graph since it is the simplest nontrivial graph whose automorphism group contains the full dihedral symmetry $\mathcal{C}_{nv}$. Thus, it can serve as a preliminary study to understand how to design symmetry-constrained control strategies. The control objective, therefore, is to design distributed controllers that drive the agents to a formation exhibiting the *full dihedral symmetry, $\mathcal{C}_{nv}$*, consisting of both rotational and reflectional relations.

Towards this goal, we proceed in stages. First, we briefly review our previous work on rotational-symmetry formation control, showing that it solves the problem for the rotational subgroup $\mathcal{C}_n$. Next, we propose an initial control strategy that enforces inter-agent reflectional relations for free reflections only. Finally, we introduce an augmented reflection-based control strategy that is able to stabilize the formation to the full $\mathcal{C}_{nv}$ symmetry.

3.1 Problem Setup

We consider a team of n agents modeled by the integrator dynamics

$$\dot{p}_i(t) = u_i(t), \quad i \in \{1, \dots, n\}, \tag{2}$$

where $p_i(t) \in \mathbb{R}^2$ is the position of agent i and $u_i(t) \in \mathbb{R}^2$ is its control to be designed. The formation control objective is to steer the state to a configuration that exhibits the full dihedral symmetry of C_n, that is, to a $\tau(\mathcal{C}_{nv})$-symmetric configuration. Formally, we want to render the set

$$\mathcal{F} = \{p \in \mathbb{R}^{2n} \,|\, \tau(\gamma)p_i = p_{\gamma(i)} \text{ for all } \gamma \in \mathcal{C}_{nv} \text{ and } i \in \mathcal{V}\},$$

asymptotically stable. Then, the objective is to satisfy

$$\lim_{t\to\infty} \left\| p_i(t) - \tau(\gamma_{ji})\, p_j(t) \right\| = 0 \tag{3}$$

for all $\gamma_{ji} \in \mathcal{C}_n \cup \mathscr{F}_s$, and

$$\lim_{t\to\infty} \left\| p_i(t) - \tau(\gamma_i) p_i(t) \right\| = 0 \tag{4}$$

for some $\gamma_i \in \mathscr{S}_s$.

We define an *interaction graph* $\mathcal{G}_I = (\mathcal{V}, \mathcal{E}_I)$ to specify which agents are able to exchange information. This graph is defined as a spanning tree subgraph of C_n which ensures the minimal connectivity required in the MAS. For example, for the $\mathcal{C}_{6v}$-symmetric framework in Fig. 1(a), the edge set $\mathcal{E}_I = \{12, 23, 34, 45, 56\}$ satisfies the connectivity requirement.

3.2 Rotation-based Formation Control

We begin by reviewing our previous work (Martinez and Zelazo (2026)) which provided a solution to the symmetry-based formation control problem for the rotational subgroup $\mathcal{C}_n$. The first step is to define a *rotational symmetry-forcing potential* over the edges in the interaction graph,

$$F_{\mathcal{C}_n}(p(t)) = \frac{1}{2} \sum_{ij \in \mathcal{E}_I} \|p_i(t) - \tau(\gamma_{ji}) p_j(t)\|^2, \tag{5}$$

where $\gamma_{ij} \in \mathcal{C}_n$. We then implement a control law defined by the gradient dynamical system

$$u(t) = -\nabla F_{\mathcal{C}_n}(p(t)), \tag{6}$$

and examine the expression of the closed-loop dynamics for each agent. For an agent i the dynamics are

$$\dot{p}_i(t) = \sum_{ij \in \mathcal{E}_I} (\tau(\gamma_{ji}) p_j(t) - p_i(t)). \tag{7}$$

Expressing the dynamics in state-space form yields the closed-loop system

$$\dot{p}(t) = -Q(\mathcal{C}_n)p(t), \tag{8}$$

where $Q(\mathcal{C}_n) \in \mathbb{R}^{2n\times 2n}$ is the resulting *symmetry-constraining* matrix-weighted Laplacian for the graph $\mathcal{G}_I$, with block entries

$$[Q(\mathcal{C}_n)]_{ij} = \begin{cases} d(i) I_2, & i = j,\ i \in \mathcal{V} \\ -\tau(\gamma_{ji}), & ij \in \mathcal{E}_I \\ 0, & \text{o.w.} \end{cases}, \tag{9}$$

where $d(i)$ denotes the degree of vertex i in $\mathcal{G}_I$, and $\gamma_{ij} \in \mathcal{C}_n$. The matrix $Q(\mathcal{C}_n)$ has the same structural form used in matrix-weighted Laplacian methods (e.g., Zhao and Zelazo (2019)). As a matrix-weighted Laplacian, it can be expressed as the matrix product $E(\mathcal{C}_n)E(\mathcal{C}_n)^T$, where $E(\mathcal{C}_n) \in \mathbb{R}^{2|\mathcal{V}|\times 2|\mathcal{E}_I|}$ results in a matrix-weighted incidence matrix structure similar to the scalar case. For an edge $k = ij$, one has $[E(\mathcal{C}_n)]_{ik} = I$ if the edge k leaves vertex i, $[E(\mathcal{C}_n)]_{ik} = -\tau(\gamma_{ji})$ if the edge k enters the vertex i, and $[E(\mathcal{C}_n)]_{ik} = 0$ otherwise.

Theorem 1 (Martinez and Zelazo (2026))**.** *Consider a MAS consisting of n integrator agents (2), whose interaction topology is defined by a spanning tree graph $\mathcal{G}_I \subset C_n$, and let*

$$\mathcal{F}_{\mathcal{C}_n} = \{p \in \mathbb{R}^{2n} | E(\mathcal{C}_n)^T p = 0,\ \ i \in \mathcal{V}\}.$$

Then, for any initial condition $p(0) \in \mathbb{R}^{2n}$, the control

$$u(t) = -Q(\mathcal{C}_n)p(t)$$

renders the set $\mathcal{F}_{\mathcal{C}_n}$ exponentially stable.

Theorem 1 establishes that the trajectories of the closed-loop system converge exponentially to the set $\mathcal{F}_{\mathcal{C}_n}$, which is precisely the set of configurations for which every enforced rotational relation is satisfied. Equivalently, $\mathcal{F}_{\mathcal{C}_n} = \text{Null}\big(Q(\mathcal{C}_n)\big)$, implying that a steady-state configuration $p^\star$ satisfies

$$E(\mathcal{C}_n)^T p^\star = 0.$$

Therefore, the control law (8) steers the agents to a unique $\mathcal{C}_n$-symmetric configuration (up to trivial rotations and scaling), but does not enforce the reflectional component of the full dihedral symmetry, potentially admitting undesired mirror-symmetric configurations.

3.3 Free Reflectional Symmetry Formation Control

Having characterized the rotational case, we now consider a similar strategy to solve the problem for reflectional symmetries. Following the same procedure as for the rotation case, we begin by focusing solely on the free (inter-agent) reflection relationships encoded by the edges. We define the *reflectional symmetry-forcing potential*

$$F_{\mathscr{F}_s}(p(t)) = \frac{1}{2}\sum_{ij\in\mathcal{E}_I} \|p_i(t) - \tau(\gamma_{ji})p_j(t)\|^2, \tag{10}$$

where $\gamma_{ij} \in \mathscr{F}_s$.

Implementing the negative gradient flow,

$$u(t) = -\nabla F_{\mathscr{F}_s}(p(t)),$$

yields the reflection-based control law

$$\dot{p}(t) = -Q(\mathscr{F}_s)\, p(t), \tag{11}$$

where $Q(\mathscr{F}_s)$ is the matrix-weighted Laplacian associated with the free reflections set $\mathscr{F}_s$. Note that this matrix has the same structure as $Q(\mathcal{C}_n)$ in (9).

We observe the following immediate properties of $Q(\mathscr{F}_s)$. First, since $Q(\mathscr{F}_s) = E(\mathscr{F}_s)E(\mathscr{F}_s)^T$, it is symmetric and positive semi-definite (PSD). Consequently,

$$p \in \text{Null}\big(Q(\mathscr{F}_s)\big) \quad\Longleftrightarrow\quad E(\mathscr{F}_s)^T p = 0.$$

We now state a result on the dimension of the null-space of $Q(\mathscr{F}_s)$.

Proposition 1. *The free reflection-based matrix-weighted Laplacian $Q(\mathscr{F}_s)$ satisfies* $\dim \text{Null}(Q(\mathscr{F}_s)) = 2$.

Proof. First, note that the condition $E(\mathscr{F}_s)^T p = 0$ encodes all reflection relations $p_i = \tau(\gamma_{ji})p_j$ associated with the edges of $\mathcal{G}_I$. Along any path between two nodes i and j, the symmetry constraints between neighboring agents yield the recursive relations

$$p_{v_{j+1}} = \tau(\gamma_{v_j\, v_{j+1}})\, p_{v_j}, \qquad j = 0, \dots, k-1,$$

where $v_0 = i$, $v_k = j$, and $v_\ell v_{\ell+1} \in \mathcal{E}_I$. Composing these relations yields the chain

$$p_j = \underbrace{\tau(\gamma_{v_{k-1}\, v_k})\tau(\gamma_{v_{k-2}\, v_{k-1}}) \cdots \tau(\gamma_{v_0, v_1})}_{S_j \in O(2)}\, p_i,$$

such that $p_j = S_j p_i$, $S_j = \tau(\gamma_{ji}) \in O(2)$, for all $j = 1, \dots, n$, with $S_i = I_2$. Hence, every configuration in the nullspace can be written as

$$p = \begin{bmatrix} p_1^T & \cdots & p_n^T \end{bmatrix}^T = \begin{bmatrix} (S_1 p_\ell)^T & \cdots & (S_n p_\ell)^T \end{bmatrix}^T,$$

for an arbitrary vertex $\ell \in \mathcal{V}$. This shows

$$\text{Null}\big(Q(\mathscr{F}_s)\big) = \big\{\, p_\ell \in \mathbb{R}^2 \big| [(S_1 p_\ell)^T \ \cdots \ (S_n p_\ell)^T]^T \big\},$$

which is a 2-dimensional subspace parameterized by $p_\ell \in \mathbb{R}^2$. Therefore, this implies $\dim \text{Null}(Q(\mathscr{F}_s)) = 2$. □

Theorem 2. *Consider a MAS consisting of n integrator agents (2), whose interaction topology is defined by a spanning tree graph $\mathcal{G}_I \subset C_n$, and let*

$$\mathcal{F}_{\mathscr{F}_s} = \{p \in \mathbb{R}^{2n} | E(\mathscr{F}_s)^T p = 0, \ \ i \in \mathcal{V}\}.$$

Then, for any initial condition $p(0) \in \mathbb{R}^{2n}$, the control

$$u(t) = -Q(\mathscr{F}_s)p(t)$$

renders the set $\mathcal{F}_{\mathscr{F}_s}$ exponentially stable.

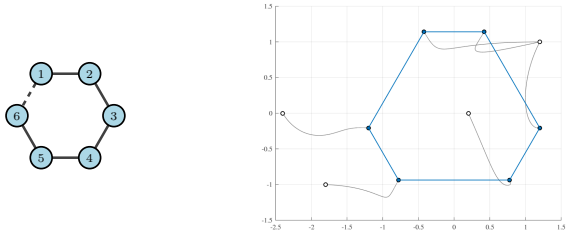
Proof. The matrix $Q(\mathscr{F}_s)$ is symmetric, PSD, and by Proposition 1 satisfies $\dim \text{Null}(Q(\mathscr{F}_s)) = 2$. Hence, the system $\dot{p}(t) = -Q(\mathscr{F}_s)p(t)$ is a linear time-invariant system whose solutions converge exponentially to $\text{Null}(Q(\mathscr{F}_s))$. Therefore, for any initial condition $p(0) \in \mathbb{R}^{2n}$, the trajectories satisfy

$$\lim_{t\to\infty} p(t) \in \text{Null}(Q(\mathscr{F}_s)) = \mathcal{F}_{\mathscr{F}_s},$$

which establishes exponential stability of $\mathcal{F}_{\mathscr{F}_s}$. □

We conclude that the subspace $\mathcal{F}_{\mathscr{F}_s}$ does not enforce the self-symmetry condition (4) since this condition applies if and only if each agent i lies on its corresponding mirror line $\mathcal{L}_i$. To show this explicitly, recall that $p \in \mathcal{F}_{\mathscr{F}_s}$ implies $p_i = S_i p_j$ for some $p_j \in \mathbb{R}^2$ and all $i \in \mathcal{V}$. Therefore, $\mathcal{F}_{\mathscr{F}_s}$ also admits configurations satisfying $p_j \notin \mathcal{L}_j$. Hence, the reflection-based control law (11) alone may drive the system to a "symmetric flexible" configuration satisfying all pairwise reflection constraints that does not coincide with the intended $\mathcal{C}_{nv}$-symmetric target formation.

Example 1. *Consider a MAS consisting of $n = 6$ agents, tasked with attaining a $\mathcal{C}_6$-symmetric configuration. The interaction graph $\mathcal{G}_I$, is defined by the edge set $\mathcal{E}_I = \{12, 23, 34, 45, 56\}$. The agents are now tasked with attaining a symmetric configuration under inter-agent free reflection constraints. The actions $\tau(\mathscr{F}_s)$ include the vertical mirror $\mathcal{L}_{12}$, which reflects agent pairs $(1, 2)$ and $(4, 5)$, as well as two diagonal mirrors: $\mathcal{L}_{23}$, reflecting $(2, 3)$ and $(5, 6)$, and $\mathcal{L}_{34}$, reflecting $(3, 4)$ and $(1, 6)$. The actions are represented by using the Householder reflection $\tau(\mathcal{L}_{ij}) = H(\hat{n}_{ij})$. Fig. 2 shows the trajectories and final configuration obtained under the control law (11).*



(a) The cycle graph C_6.

(b) Agents converge to a $\tau(\mathcal{C}_n)$-symmetric formation.

Fig. 2. Simulation results for Example 1.

In Example 1, observe that control law (11) drives the agents to a configuration that satisfies the prescribed pairwise reflection relations but does not enforce the full dihedral symmetry of the desired C_{6v}-symmetric configuration.

3.4 Full $\mathcal{C}_{nv}$ Symmetry Formation Control

We now propose an augmentation of the reflection-based control law so that the closed-loop dynamics enforce all

symmetry relations in $\gamma \in \mathscr{R}_s = \mathscr{F}_s \cup \mathscr{S}_s$. As a key result in this work we show that enforcing all free symmetries in addition to a single designated agent ℓ anchored to its prescribed mirror line $\mathcal{L}_\ell$ suffices to satisfy all relations $\gamma \in \mathcal{C}_{nv}$ of a $\mathcal{C}_{nv}$-symmetric configuration.

We augment the potential (5) with the term

$$F_\ell = \frac{1}{4}\|p_\ell(t) - \tau(\gamma_\ell)p_\ell(t)\|^2, \tag{12}$$

for some $\gamma_\ell \in \mathscr{S}_s$. We now define the augmented gradient control law

$$\begin{aligned} u(t) =& -\nabla(F(p(t)) + F_\ell(p(t))) \\ =& -Q(\mathscr{F}_s)p(t) - \mathrm{diag}(\mathrm{e}_\ell) \otimes (I_2 - \tau(\gamma_\ell))p(t). \end{aligned} \tag{13}$$

This yields the following augmented matrix-weighted Laplacian

$$\mathcal{Q}(\mathscr{R}_s) = Q(\mathscr{F}_s) + \mathrm{diag}(\mathrm{e}_\ell) \otimes (I_2 - \tau(\gamma_\ell)), \tag{14}$$

where $Q(\mathscr{F}_s)$ is the matrix-weighted Laplacian associated with the free reflection set $\mathscr{F}_s$.

Intuitively, since the interaction graph is a spanning tree, the single constraint $\gamma_\ell \in \mathscr{S}_s$ propagates through the reflection relations to all agents.

Lemma 1. *Let $\mathcal{G}_I \subset C_n$ be a spanning tree. For each edge $ij \in \mathcal{E}_I$, let $\gamma_{ji} \in \mathscr{F}_s$ be the free reflection between agents i and j. Let $\ell \in \mathcal{V}$ be a designated node constrained to the mirror line $\mathcal{L}_\ell \subset \mathbb{R}^2$ corresponding to a self-reflection $\gamma_\ell \in \mathscr{S}_s$. Then there exists a unique assignment of mirror lines $\mathcal{L}_i$ for each $i \in \mathcal{V}$, such that for every edge $ij \in \mathcal{E}_I$ the mirror lines satisfy $\mathcal{L}_j = \tau(\gamma_{ji})\mathcal{L}_i$. In particular, the orientation of every agents' mirror axis $\mathcal{L}_i$ is uniquily determined by $\mathcal{L}_\ell$ and the inter-agent reflection symmetries.*

Proof. Since $\mathcal{G}_I$ is a spanning tree, there is a unique path from ℓ to any node i. Along this path define

$$S_i := \tau(\gamma_{v_k v_{k-1}})\tau(\gamma_{v_{k-1}v_{k-2}}) \cdots \tau(\gamma_{v_1 v_0}) \in O(2),$$

with $S_\ell = I_2$, $v_0 = \ell$ and $v_k = i$. We now *define* the mirror line of agent i by

$$\mathcal{L}_i := S_i \mathcal{L}_\ell.$$

This assignment satisfies the required inter-agent relation because, if ij is an edge and j is farther from ℓ than i, then by construction $S_j = \tau(\gamma_{ji})\, S_i$, and therefore

$$\mathcal{L}_j = S_j \mathcal{L}_\ell = \tau(\gamma_{ji})\mathcal{L}_i.$$

Thus a valid assignment exists.

To establish uniqueness, let $\{\mathcal{L}'_i\}_{i=1}^{|\mathcal{V}|}$ be any other assignment with $\mathcal{L}'_\ell = \mathcal{L}_\ell$ and $\mathcal{L}'_j = \tau(\gamma_{ji})\mathcal{L}'_i$ for all edges $ij \in E_I$. Since every agent is reached from ℓ by a unique path, an induction along that path gives

$$\mathcal{L}'_i = \tau(\gamma_{v_i v_{i-1}}) \cdots \tau(\gamma_{v_1 v_0})\mathcal{L}'_\ell = S_i \mathcal{L}_\ell = \mathcal{L}_i.$$

Hence $\mathcal{L}'_i = \mathcal{L}_i$ for all i, proving uniqueness. □

We now consider the augmented reflection-based control law

$$\dot{p}(t) = -\mathcal{Q}(\mathscr{R}_s)\, p(t). \tag{15}$$

The properties of the matrix $\mathcal{Q}(\mathscr{R}_s)$ are of interest. First, note that $I_2 - \tau(\gamma_\ell) = 2\hat{n}_\ell \hat{n}_\ell^T$. Since

$$\mathcal{Q}(\mathscr{R}_s) = E(\mathscr{F}_s)E(\mathscr{F}_s)^T + \mathrm{diag}(\mathrm{e}_\ell) \otimes 2\hat{n}_\ell \hat{n}_\ell^T,$$

for any $p \in \mathbb{R}^{2n}$ we obtain

$$\begin{aligned} p^T \mathcal{Q}(\mathscr{R}_s)p =& p^T \left(E(\mathscr{F}_s)E(\mathscr{F}_s)^T + \mathrm{diag}(\mathrm{e}_\ell) \otimes 2\hat{n}_\ell \hat{n}_\ell^T\right) p \\ =& \|E(\mathscr{F}_s)^T p\|^2 + 2|\hat{n}_\ell^T p_\ell|^2 \geq 0. \end{aligned}$$

Hence, $\mathcal{Q}(\mathscr{R}_s)$ is PSD and symmetric. Additionally, for any $p \in \mathbb{R}^{2n}$ we have

$$p^T \mathcal{Q}(\mathscr{R}_s)p = 0 \iff \begin{cases} E(\mathscr{F}_s)^T p & = 0 \\ (I_2 - \tau(\gamma_\ell))p_\ell & = 0 \end{cases}.$$

Proposition 2. *The augmented reflection-based matrix-weighted Laplacian $\mathcal{Q}(\mathscr{R}_s)$ satisfies* $\dim \mathrm{Null}(Q(\mathscr{R}_s)) = 1$.

Proof. The proof follows similarly to Proposition 1 and is omitted due to space constraints. □

We now show that the control law drives the agents from any initial condition to a desired $\mathcal{C}_{nv}$-symmetric configuration.

Theorem 3. *Consider a MAS consisting of n integrator agents (2), whose interaction topology is defined by a spanning tree graph $\mathcal{G}_I \subset C_n$, and let*

$$\mathcal{F} = \{p \in \mathbb{R}^{2n} \,|\, \tau(\gamma)p_i = p_{\gamma(i)} \text{ for all } \gamma \in \mathcal{C}_{nv} \text{ and } i \in \mathcal{V}\}.$$

Then, for any initial condition $p(0) \in \mathbb{R}^{2n}$, the control (15) renders the set $\mathcal{F}$ exponentially stable with $p(\infty)$ as the orthogonal projection of $p(0)$ onto $\mathcal{F}$,

$$\lim_{t\to\infty} p(t) = \frac{1}{n} V_0 V_0^T p(0), \tag{16}$$

where

$$V_0 = \left[(S_1\hat{\mathcal{L}}_\ell)^T \ \cdots \ (S_n\hat{\mathcal{L}}_\ell)^T\right]^T \in \mathbb{R}^{2n},$$

$\hat{\mathcal{L}}_\ell$ is a unit direction vector along the mirror line $\mathcal{L}_\ell$, and S_i are defined in Proposition 1. Furthermore, the steady-state of each agent is defined as

$$\lim_{t\to\infty} p_i(t) = \frac{1}{n} S_i \hat{\mathcal{L}}_\ell \hat{\mathcal{L}}_\ell^T \sum_{k=1}^{n} S_k^T p_k(0). \tag{17}$$

Proof. From Proposition 2 we have

$$\mathrm{Null}\big(\mathcal{Q}(\mathscr{R}_s)\big) = \mathrm{IM}(V_0),$$

Then we have

$$V_0^T V_0 = \sum_{i=1}^{n} |S_i \hat{\mathcal{L}}_\ell|^2 = \sum_i 1 = n.$$

Since $\mathcal{Q}(\mathscr{R}_s)$ is PSD and the columns of V_0 are orthogonal, we can define $\hat{V}$ as an orthonormal eigenbasis $\hat{V} = \begin{bmatrix}\hat{V}_0 & \hat{V}_+\end{bmatrix}$ with $\hat{V}_0 = \frac{1}{\sqrt{n}} V_0$ and V_+ as the orthogonal complement of V_0. The proof continues in the same way as in Theorem 2, concluding with

$$\lim_{t\to\infty} p(t) = \hat{V}_0 \hat{V}_0^T p(0) = \frac{1}{n} V_0 V_0^T p(0)\ ,$$

which shows (16). Also note that

$$V_0^T p(0) = \sum_{k=1}^{n} (S_k \hat{\mathcal{L}}_\ell)^T p_k(0) = \sum_{k=1}^{n} \hat{\mathcal{L}}_\ell^T S_k^T p_k(0).$$

From (16), the i-th agent's component is therefore

$$\lim_{t\to\infty} p_i(t) = \frac{1}{n} S_i \hat{\mathcal{L}}_\ell \hat{\mathcal{L}}_\ell^T \sum_{k=1}^{n} S_k^T p_k(0),$$

which shows (17).

We now show that $p(\infty)$ lies in the set $\mathcal{F}$. From (16) we have

$$\alpha = \frac{1}{n} V_0^T p(0) \in \mathbb{R},$$

such that

$$p(\infty) = \alpha V_0 \in \mathrm{IM}(V_0), \quad p_i(\infty) = \alpha S_i \hat{\mathcal{L}}_\ell, \quad i \in \mathcal{V}. \quad (18)$$

We are required to verify that

$$(I_2 - \tau_\ell) p_\ell = 0 \quad \Longleftrightarrow \quad (I_2 - \tau_i) p_i = 0,$$

for all $i \in \mathcal{V}$. By definition, each agent's mirror line satisfies

$$\mathcal{L}_i = S_i \mathcal{L}_\ell.$$

Therefore, since $p_i = S_i p_\ell$ and $p_\ell \in \mathcal{L}_\ell$. It follows that

$$p_i \in S_i \mathcal{L}_\ell = \mathcal{L}_i \quad \Longleftrightarrow \quad (I_2 - \tau_i) p_i = 0.$$

So we have

$$E(\mathscr{R}_s)^T p(\infty) = 0, \quad (I_2 - \tau(\gamma_i))\, p_i(\infty) = 0.$$

Consequently, $p(\infty)$ satisfies all enforced reflectional relations and self-symmetry constraints.

$$p(\infty) \in \mathrm{Null}\big(Q(\mathscr{R}_s)\big).$$

Additionally, we show that the steady-state configuration $p(\infty)$ is invariant under every action in $\mathcal{C}_n$. This is expressed by the condition

$$\tau(\rho)\, p_i(\infty) = p_{\rho(i)}(\infty), \quad \forall\, i \in \mathcal{V},\ \rho \in \mathcal{C}_n.$$

To verify this identity, we use the group property

$$\gamma_{\rho(i)\ell} = \rho \gamma_{i\ell},$$

yielding

$$p_{\rho(i)}(\infty) = \tau(\gamma_{\rho(i)\ell}) p_\ell = \tau(\rho)\, \tau(\gamma_{i\ell}) p_\ell = \tau(\rho)\, p_i(\infty),$$

which satisfies the rotational symmetry condition of a $\mathcal{C}_n$-symmetric configuration. This equivalently means $E(\mathcal{C}_n)^T p(\infty) = 0$, hence, $p(\infty) \in \mathrm{Null}\big(Q(\mathcal{C}_n)\big)$.

Finally, by definition, $\mathcal{F} = \mathrm{Null}\big(Q(\mathcal{C}_n)\big) \cap \mathrm{Null}\big(Q(\mathscr{R}_s)\big)$. The relations established above for all $\gamma \in \mathcal{C}_n$ and for all $\gamma \in \mathscr{R}_s$ imply that

$$\tau(\gamma) p_i(\infty) = p_{\gamma(i)}(\infty) \quad \text{for all } \gamma \in \mathcal{C}_{nv},\ i \in \mathcal{V}.$$

Equivalently, $p(\infty) \in F$. Therefore, Since $p(t)$ converges exponentially to $p(\infty)$, it follows that the set F is exponentially stable, as claimed. □

Example 2. *Consider the same setup as in Example 1, now are tasked with attaining a $\mathcal{C}_{nv}$-symmetric configuration solely under reflection constraints. Fig. 3 shows the resulting trajectories and final configuration as a $\tau(\mathcal{C}_{nv})$-symmetric framework obtained with the augmented control law* (15).

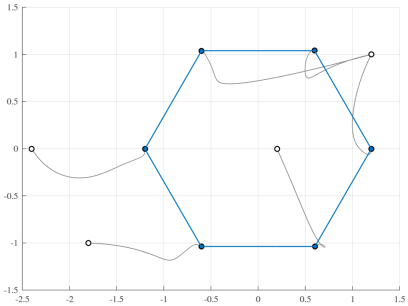


Fig. 3. Simulation results for Example 2.

4. CONCLUSION

This paper introduced a symmetry-based formation control method in $\mathbb{R}^2$ that relies solely on inter-agent reflection relations together with a single anchored agent. We showed that any $\mathcal{C}_{nv}$-symmetric formation can be achieved using only $n-1$ communication links, with guaranteed stability and convergence. These results highlight a potential alternative to rigidity-based formation control, and future directions include extending the framework to point group symmetries in $\mathbb{R}^3$, analyzing more general information structures such as directed and switching graphs, and incorporating leader–follower structures to address the formation maneuvering problem.

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